Lovro Vrcek

Postdoctoral Researcher

vrcek.lovro@gmail.com
github.com/lvrcek
linkedin/in/lovro-vrcek
Google Scholar URL

SUMMARY

I am a postdoctoral researcher at the intersection of AI and genomics, with a background in physics and a broader interest in AI-driven approaches to scientific discovery. My work focuses on geometric deep learning and foundation models applied to genome assembly and genomics more broadly. I am particularly interested in understanding when and why AI methods succeed or fail in biological contexts, and in how AI can contribute to other domains, including drug discovery and materials design.

WORK EXPERIENCE

Genome Institute of Singapore, A*STAR, Singapore
Postdoctoral Researcher — Jun 2025 – present

- Demonstrated that genomic foundation models behave fundamentally differently from text LLMs via entropy analysis, questioning the value of DNA pretraining; will be presented at ICLR - Learning Meaningful Representations of Life workshop 2026 [arXiv]

Genome Institute of Singapore, A*STAR, Singapore
Senior Research Officer — Oct 2021 – Jun 2025

- Developed GNNome, a GNN-based de novo genome assembler representing a new paradigm for the field; achieved 20% improvement in NG50 and NGA50 over SOTA on human genome with comparable results across three other species; published in Genome Research 2025
- Contributed to the first de novo assembly of an Indian reference genome, providing a valuable genomic resource for the South Asian population [bioRxiv]
- Lectured at the ASTAR/NTU AI + Computational Genomics course covering deep learning and genome assembly

Genome Institute of Singapore, A*STAR, Singapore
Intern (ARAP Scholarship Awardee) — Oct 2019 – Oct 2021

- Pioneered neural algorithmic reasoning approaches to de novo genome assembly; presented at NeurIPS - Learning Meets Combinatorial Algorithms workshop 2020 [arXiv]
- Developed a CNN-based classifier for long read type classification, achieving competitive accuracy across multiple classes

Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia
Research and Teaching Assistant — Oct 2018 – Oct 2019

- Taught introductory Programming and Data Science courses

EDUCATION

Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia
PhD in Computer Science (GPA: 4.86 / 5.00) — Oct 2018 – Jun 2025

- Thesis: De novo genome assembly based on graph neural networks
- Supervisors: Mile Sikic and Xavier Bresson
- Graduated with high honour - Magna cum laude

Faculty of Electrical Engineering and Computing, University of Zagreb, Croatia
BSc in Computer Science (GPA: 4.60 / 5.00) — Oct 2016 – Jul 2019

- Thesis: DNA sequence polishing using deep learning
- Supervisor: Mile Sikic

Faculty of Science, University of Zagreb, Croatia
Integrated BSc and MSc in Research Physics (GPA: 4.96 / 5.00) — Oct 2013 – Jul 2018

- Thesis: Machine learning in solid-state physics and statistical physics
- Supervisors: Vinko Zlatic and Ivor Loncaric
- Graduated with highest honour - Summa cum laude

Note: Grades vary from 1 to 5, where 5 is the highest possible grade

MAJOR ACHIEVEMENTS

ARAP Scholarship	A*STAR attachment programme, 2019.
Rector's award UNIZG	Individual research, 2018.

TECHNICAL SKILLS

Programming Languages	Python, Go, R, Bash, C, C++
ML Frameworks	PyTorch, Huggingface, PyG/DGL, Scikit-learn, Numpy, Matplotlib
Bioinformatics	samtools, minimap2, IGV, hifiasm
Other	Linux, Git, LaTeX, HTML, CSS

ORGANIZATIONAL SKILLS

- **AI4NA Workshop**: One of the organizers of AI for Nucleic Acids workshop at ICLR 2025
- **GIS Young Researcher Network**: Organized professional development events for early career researchers
- **STEM Games 2019**: Organized Technology Arena and authored algorithmic programming problems for contestants

INVITED TALKS

- **Starkly Speaking, Valence Labs (Sep 2024 - online)**:
  Geometric deep learning framework for de novo genome assembly [YouTube]
- **Drug discovery journal club, Johannes Kepler University Linz (Mar 2024 - online)**:
  Geometric deep learning framework for de novo genome assembly
- **Computer Laboratory, University of Cambridge (Jun 2023)**:
  Untangling genome assembly graphs with graph neural networks [YouTube]

PUBLICATIONS

- **Vrček, L.**, Bresson, X., Laurent, T., Schmitz, M., Šikić, M., *Geometric deep learning framework for de novo genome assembly*, Genome Research, gr. 279307.124, 2025.
- **Vrček, L.**, Bresson, X., Laurent, T., Schmitz, M., Šikić, M., *Untangling Genome Assembly Graphs with Graph Neural Networks*, ICML - Workshop on Computational Biology, 2023. (non-archival)
- Lindner, S., Lončarić, I., **Vrček, L.**, Alducin, M., Juaristi, J. I., Saalfrank, P., *Femtosecond Laser-Induced Desorption of Hydrogen Molecules from Ru(0001): A Systematic Study Based on Machine-Learned Potentials*, J. Phys. Chem. C, 2023, 127, 30, 14756–14764
- **Vrček, L.**, Veličković, P., Šikić, M., *A step towards a neural genome assembly*, NeurIPS - Learning Meets Combinatorial Algorithms workshop, 2020.
- **Vrček, L.** Huang, M. H. H., Vaser, R., Šikić, M., *Deep learning apporach to determining the type of long reads*, poster presentation, International Conference on Intelligent Systems for Molecular Biology, 2020.
- **Vrček, L.**, Šikić, M., *Supervised learning approach to long read classification*, poster presentation, Fourth International Workshop on Data Science, 2019.

PREPRINTS

- Rochkoulets, M., **Vrček, L.**, Šikić, M., *Entropy, Disagreement, and the Limits of Foundation Models in Genomics* , arXiv preprint, 2026.
- Sarashetti, P., et al. *A Complete Telomere-to-Telomere Diploid Reference Genome for South Asian Population* , bioRxiv preprint, 2025.
- Schmitz, M., **Vrček, L.**, Kawaguchi K., Šikić, M., *DipGNNome: Diploid de novo genome assembly with geometric deep learning and beam-search* , bioRxiv preprint, 2025.

OTHER SKILLS AND INTERESTS

Languages	English (fluent), Croatian (native), German (basic)
Interests	Rock climbing, Hiking, Weightlifting, Swimming